

Play-by-Play Modelling with nba_api



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Tools and Stack

Language: Python

Data: nba_api

Key Libraries: polars, SciPy, statsmodels, scikit-learn

Visualizations: Plotly, Seaborn, Matplotlib

Key Deliverables

Clean event dataset

Transition matrix

Markov baseline model

ML model + metrics

Visual analysis notebook

Project Scope

Play-by-play ingestion

Event sequence modeling

Markov baseline + ML classifier

Multi-season support



Project Brief

This project aims to effectively use NBA's play-by-play data using the [nba_api](#) to build a statistical model that leverages and forecasts in-game events over time.

- Pull play-by-play logs for many games
- Engineer state variables that describe the game at each event (time, score, players etc..)
- Model what happens next (shot, turnover, foul, rebound) as a function of the current game state using Machine Learning.

Core Research Question

Given the current game state at a play-by-play event, can we predict - *What is the next event type in the sequence* (Ex: Field Goal Attempt, Turnover, Foul, Rebound, Timeout)

Initial Methodology

1. Data Collection

- Fetch game IDs from the last 10 years
- For each game, fetch play-by-play data
- Save/Cache raw responses

2. Pre-processing and Feature Engineering

- Clean and normalize event data (event/types, descriptions)
- Structure data by:
 - Game clock & quarter
 - Score differential at event time
 - Possession team and offensive/defensive team
- Create a "next event type" column as the prediction target

3. Modelling Approaches

- Set a baseline Markov Model
 - Build a transition matrix between event types
 - Given current event type, predict next type through empirical probabilities
- Supervised ML Model
 - Set features (ex: current event type, score difference, time remaining, quarter)
 - Implement 2-3 apt models (ex: Logistic Regression, Random Forest or Gradient Boosting)

4. Evaluation and Interpretation

- Generate metrics such as Accuracy, F1, and Confusion matrix across event types
- Compare Markov baseline model to ML model
- Identify and analyze model errors i.e. where most predictions fail